

Unit 6 Wiki Entry 1: Future Direction of Security and Risk Management

Issue Selected: Automated Cyber Defence and AI-Driven Security

A key factor shaping the future of security and risk management is the increasing use of **artificial intelligence (AI) in automated cyber defence systems**. As cyber threats become faster and more complex, traditional manual detection methods are becoming less effective, leading organisations to adopt AI-driven security solutions.

Vyas et al. (2023) explain that automated cyber defence systems can detect and respond to threats in real time using machine learning and behavioural analysis. This improves incident response speed and reduces reliance on human intervention, especially in large-scale or high-volume attack environments.

However, AI-based security also introduces new risks. Khan and Ghafoor (2024) highlight **adversarial machine learning**, where attackers manipulate inputs or exploit weaknesses in AI models to bypass detection systems. This creates a situation where defensive systems themselves become attack targets.

From a broader risk perspective, Aven (2016) and Zio (2018) argue that modern risk management must move beyond static models and incorporate uncertainty, complexity, and real-time system behaviour. This is particularly relevant in AI-driven environments where threats constantly evolve.

Overall, the future of security and risk management is likely to focus on **adaptive, AI-augmented defence systems**, supported by governance frameworks such as ISO/IEC 27001 and NIST CSF. However, balancing automation with human oversight will remain essential to manage emerging risks.

References

Aven, T. (2016) *European Journal of Operational Research*, 253(1), pp. 1–13.
Khan, M. and Ghafoor, L. (2024) *Journal of Computational Intelligence and Robotics*, 4(1), pp. 51–63.
Vyas, S. et al. (2023) *Analytical and Computational Systems*, 37(4), pp. 111–142.
Zio, E. (2018) *Reliability Engineering & System Safety*, 177, pp. 176–190.

> [Comments \(0\)](#)



Unit 6 Wiki Entry 2: Risk Management, Ethics, and Emerging Security Challenges

Issue Selected: Evolving Risk Management in Complex Digital Systems

Modern risk management is becoming increasingly complex due to rapid digital transformation, interconnected systems, and evolving cyber threats. According to Aven (2016), traditional risk assessment models are often limited because they assume static conditions, whereas modern systems are dynamic and uncertain.

In this context, organisations must adopt more flexible and adaptive risk management approaches. Zio (2018) highlights the importance of using advanced analytics and real-time data to improve risk understanding in complex systems, particularly where uncertainty is high.

A further challenge is the increasing integration of automated and AI-driven systems into security operations. While these systems improve efficiency, they also raise ethical and operational concerns, such as accountability, transparency, and decision-making control. Khan and Ghafoor (2024) emphasise that adversarial manipulation of AI systems can undermine trust in automated security tools.

Additionally, Vyas et al. (2023) suggest that automated cyber defence systems must still include human oversight to ensure correct interpretation of alerts and avoid over-reliance on automation.

Overall, future risk management will require a combination of **technical controls, governance frameworks, and ethical oversight**. Standards such as ISO/IEC 27001 and NIST CSF will remain important, but they must evolve to support AI-driven environments and increasingly complex digital ecosystems.

References

Aven, T. (2016) *European Journal of Operational Research*, 253(1), pp. 1–13.
Khan, M. and Ghafoor, L. (2024) *Journal of Computational Intelligence and Robotics*, 4(1), pp. 51–63.
Vyas, S. et al. (2023) *Analytical and Computational Systems*, 37(4), pp. 111–142.
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